

Kunming AI+

An Applied Introduction to AI with PyTorch

Luca Ghafourpour



**UNIVERSITY OF
CAMBRIDGE**
Department of Engineering

Session Title	Topics
Workshop 1	VS Code, Conda and PyTorch
Practical Workshop 1	Constructing and training a neural network from scratch
Workshop 2	PyTorch modules for building and training neural networks
Practical Workshop 2	PyTorch modules for data loading
Workshop 3	Diagnosing training performance + <i>Weights and Biases</i>
Practical Workshop 3	Initialisation and regularisation
Academic Lecture 1	Physics Informed Neural Networks
Academic Lecture 2	Neural Operators
Workshop 4	Convolutional neural networks
Practical Workshop 4	Scikit Learn
Workshop 5	Complete end-to-end experiment
Practical Workshop 5	Open Q&A

From running training to reading it

Last session we finished abstracting the original training loop we built from scratch by introducing PyTorch's Dataset and DataLoader.

Today we will learn how to read the train and validation loss curves to diagnose six common problems:

1

Healthy baseline

2

Underfitting

3

Overfitting

4

Bias–variance

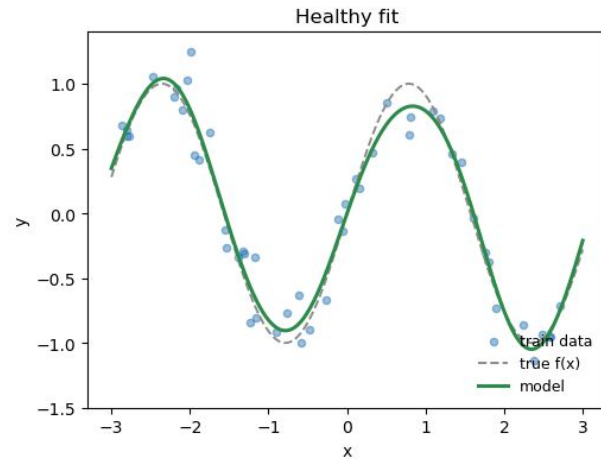
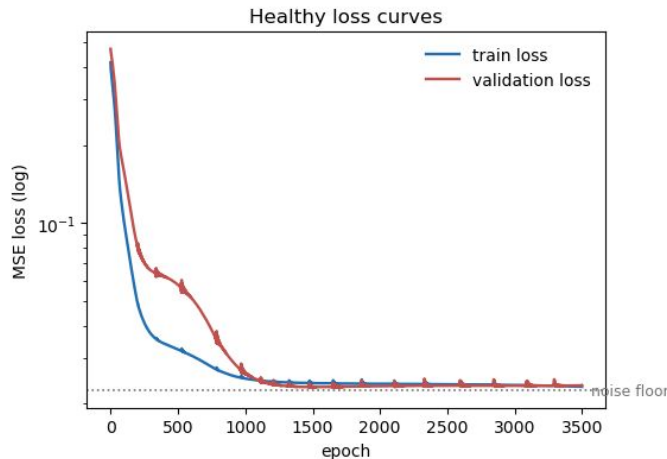
5

Learning rate

6

Batch size

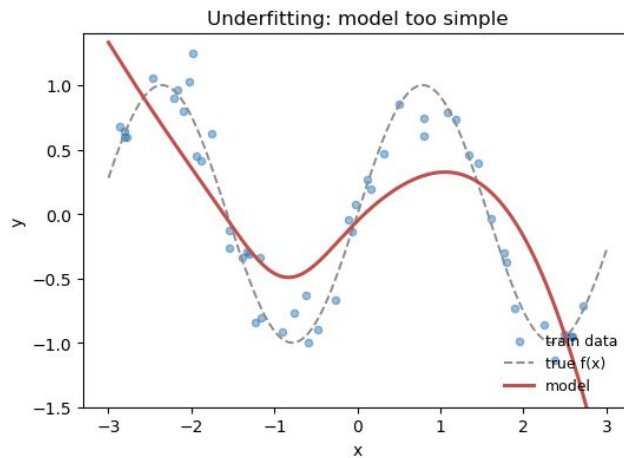
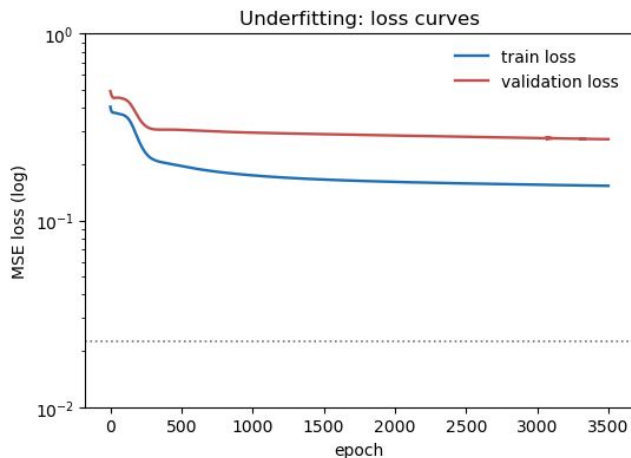
Train vs Validation Loss Curves



Both curves descend and flatten with a small stable gap. The validation rests on the noise floor.

The fit tracks the trend and ignores the noise.

Underfitting



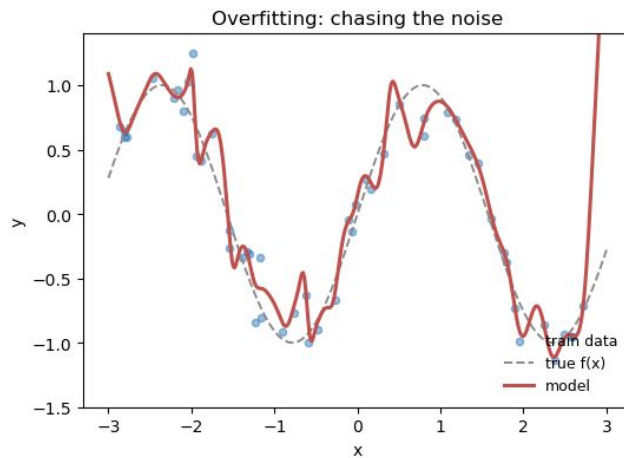
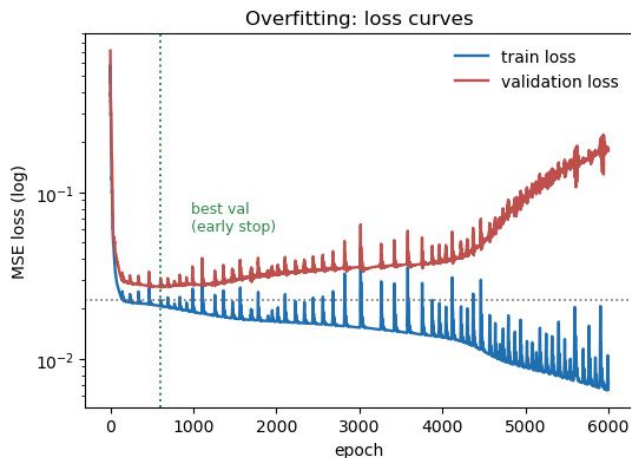
OBSERVATION

Training and validation loss are both high and flat. The model cannot even fit the training data.

POSSIBLE FIX

Increase model capacity (wider / deeper), train for more epochs, reduce regularisation, or raise the learning rate.

Overfitting



OBSERVATION

Training loss keeps falling, even below the noise floor. Validation bottoms out, then begins to grow. The model is memorising the training data.

POSSIBLE FIX

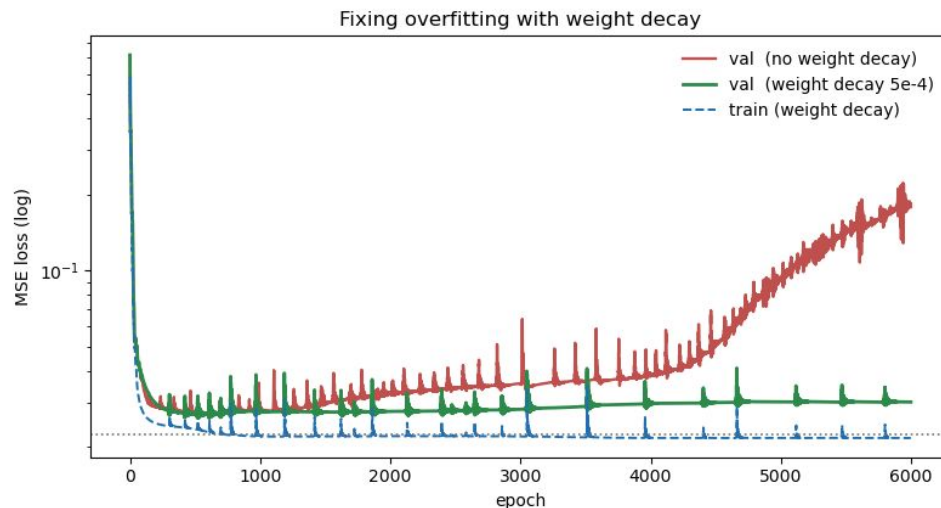
Stop early (dotted line), add weight decay or dropout, get more data, or reduce the model capacity (less wide / deep).

Weight Decay as A Fix to Overfitting

Penalises large parameter values by adding a regularisation term to the loss function.

Encourages simpler, smoother models that are less likely to memorise training examples.

Reduces overfitting and improves performance on unseen data.

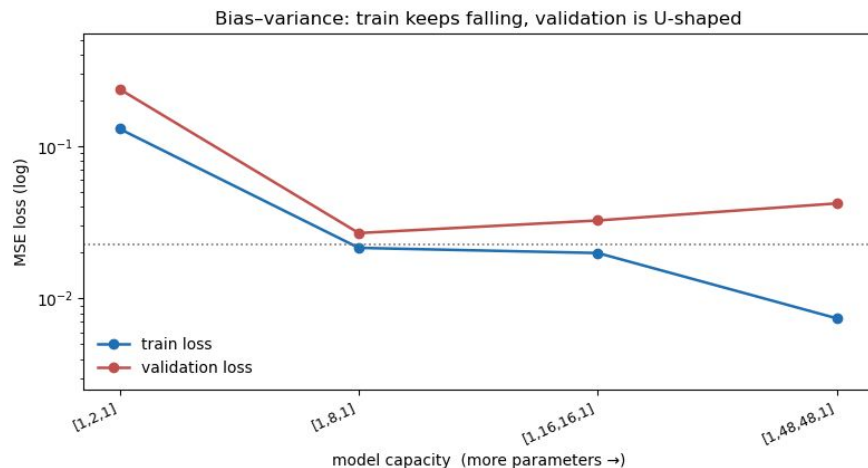


Capacity and the bias–variance trade-off

Underfitting (high bias): Training and validation losses remain high and similar. The model is too simple to capture the underlying trend.

Overfitting (high variance): Training loss continues to decrease while validation loss increases. The model is memorising the training data rather than generalising.

Good generalisation: Achieved at the balance between bias and variance, typically near the minimum of the validation loss curve before overfitting begins.

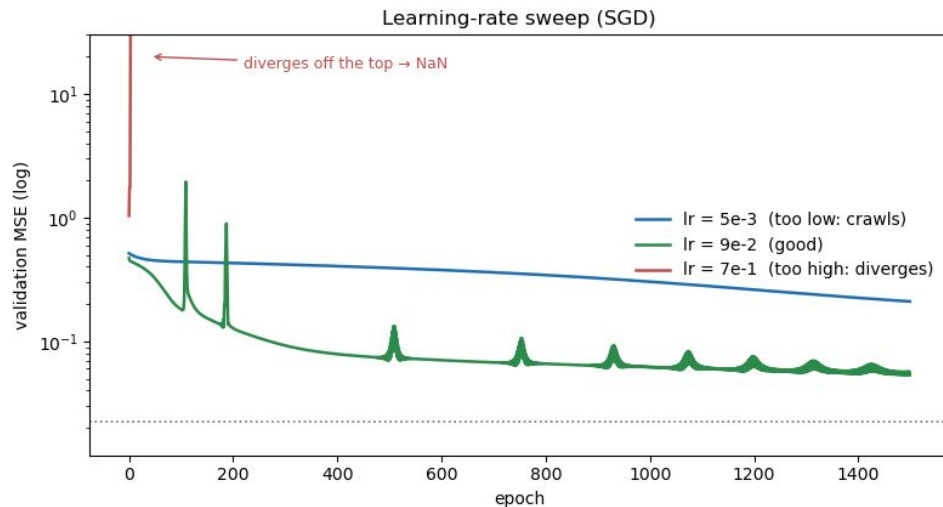


Learning rate

If the loss curve falls very slowly, never reaching the floor, increase the learning rate.

If the loss curves are spiking, or diverging to infinity, decrease the learning rate.

A good loss curve should be smooth, with descent to the floor.

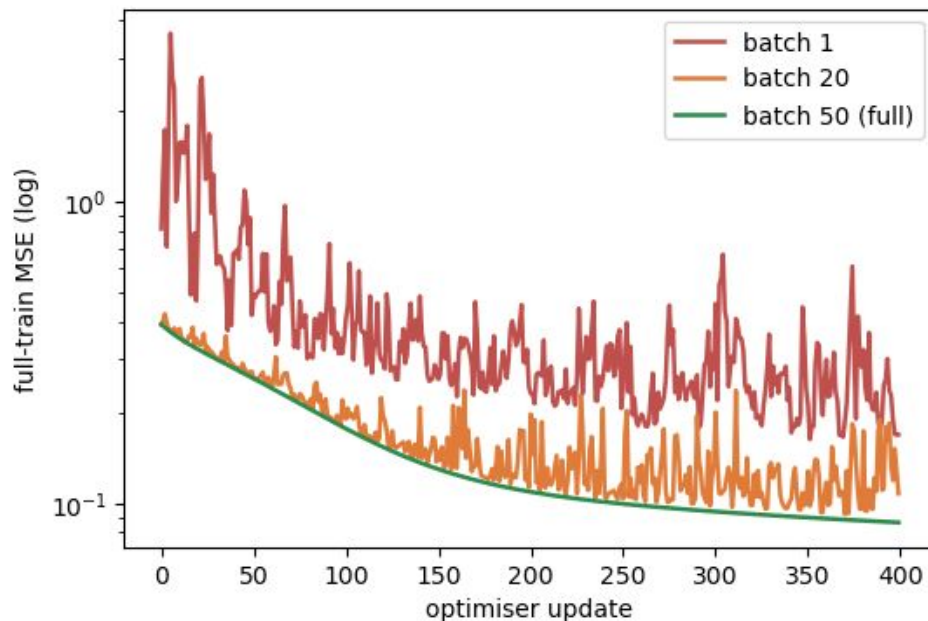


Batch size and gradient noise

A smaller batch gives a noisier gradient, and noisy loss curves. Full batch averages over all the data giving a smooth loss curve.

Trade-off: small batches result in more parameter updates per epoch. Large batches make less updates.

Fix: Either use larger batch, or a lower learning rate to match.



A diagnostic cheat-sheet

What you see in the curves	Likely cause	What to try
Both high and flat with a tiny gap across epochs	Underfitting	Bigger model, train longer, less regularisation
Train loss is very low, validation loss decreases but then increases	Overfitting	Early stopping, weight decay, reduce model capacity, add more training data
Loss falls very slowly, never converges	Learning rate too low	Increase the learning rate, use Adam
Loss spikes / oscillates, or jumps to NaN	Learning rate too high	Lower the learning rate, clip gradients
Per-update loss is jagged / very noisy	Batch size too small	Larger batch, lower learning rate

Time for some code...



Weights & Biases Tutorial

Recap

- 1 Always plot train and validation together, on a log scale — the gap between them is where the diagnosis lives.
- 2 Healthy = both descend and flatten; small, stable gap; validation near the noise floor.
- 3 Underfitting vs overfitting are the two ends of the bias–variance trade-off, set by capacity.
- 4 Learning rate has four signatures: too low, good, too high, and diverging.
- 5 Batch size too small shows up as a jagged, noisy curve.
- 6 How to use Weights & Biases to keep track of experiments